



Sales Analysis Project – Final Report



1. Project Overview

This project analyzes **regional sales performance across the U.S.** using historical data to uncover:

- Revenue trends
- Profitability drivers
- Customer & product performance
- Regional & channel insights



Tools Used:

- **Python (EDA & Visualization)**
 - **Power BI (Dashboarding)**
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2. Data Understanding

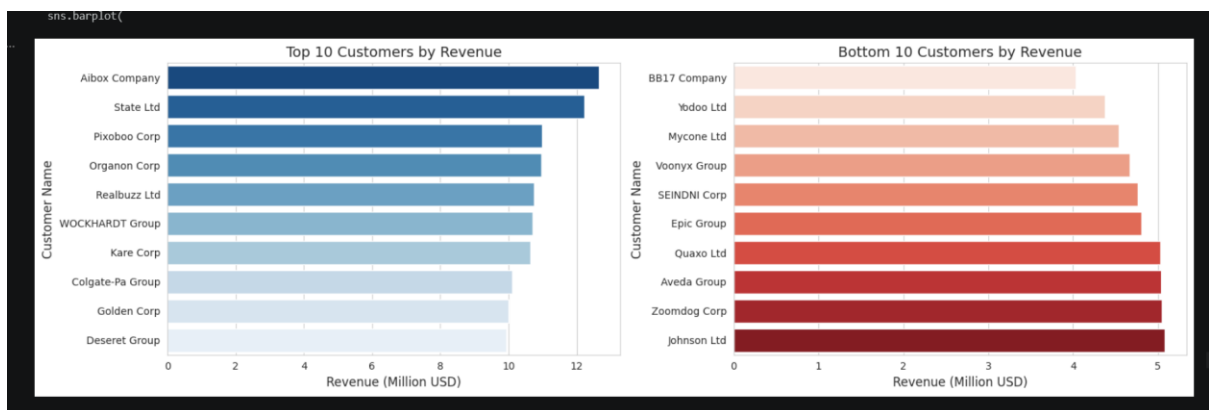
Dataset Includes:

- **Financial Metrics:** Revenue, Cost, Profit, Profit Margin
- **Customer Info:** Customer Name, Orders
- **Product Info:** Product Name
- **Geography:** State, Region
- **Time:** Month, Year
- **Channel:** Wholesale, Distributor, Export

✓ Data was clean (no missing values, no duplicates)

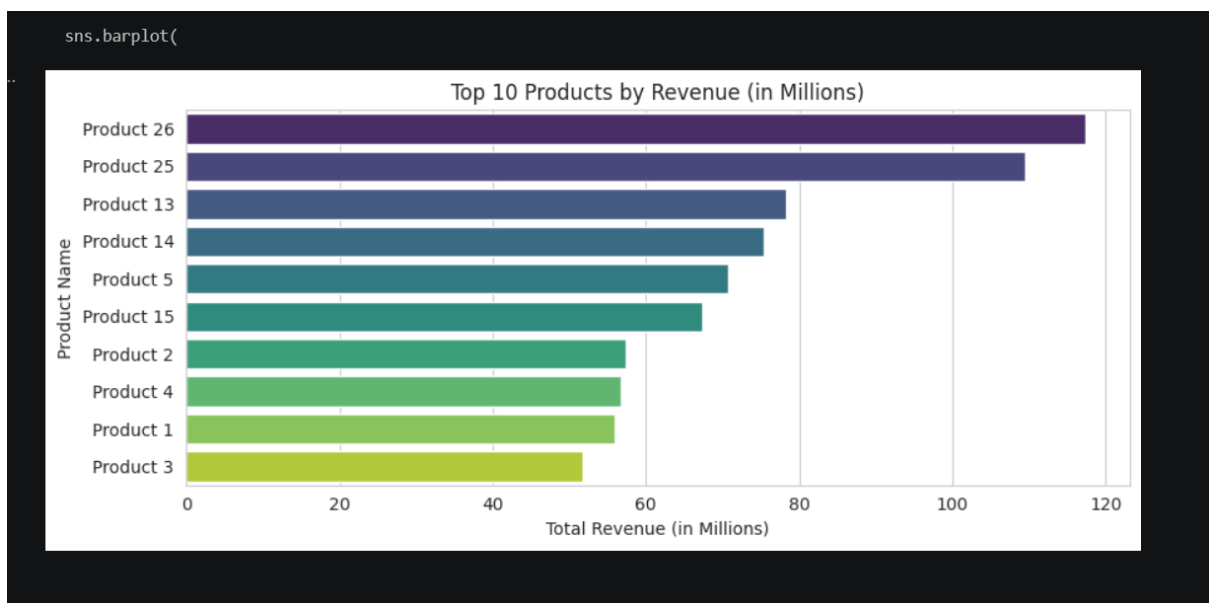
🔍 3. Exploratory Data Analysis (EDA)

📌 Customer Revenue Analysis



- Top customers (like **Aibox Company**) dominate revenue
- Bottom customers contribute significantly less (~\$4–5M)
- 👉 **Revenue is highly concentrated**

📌 Product Performance

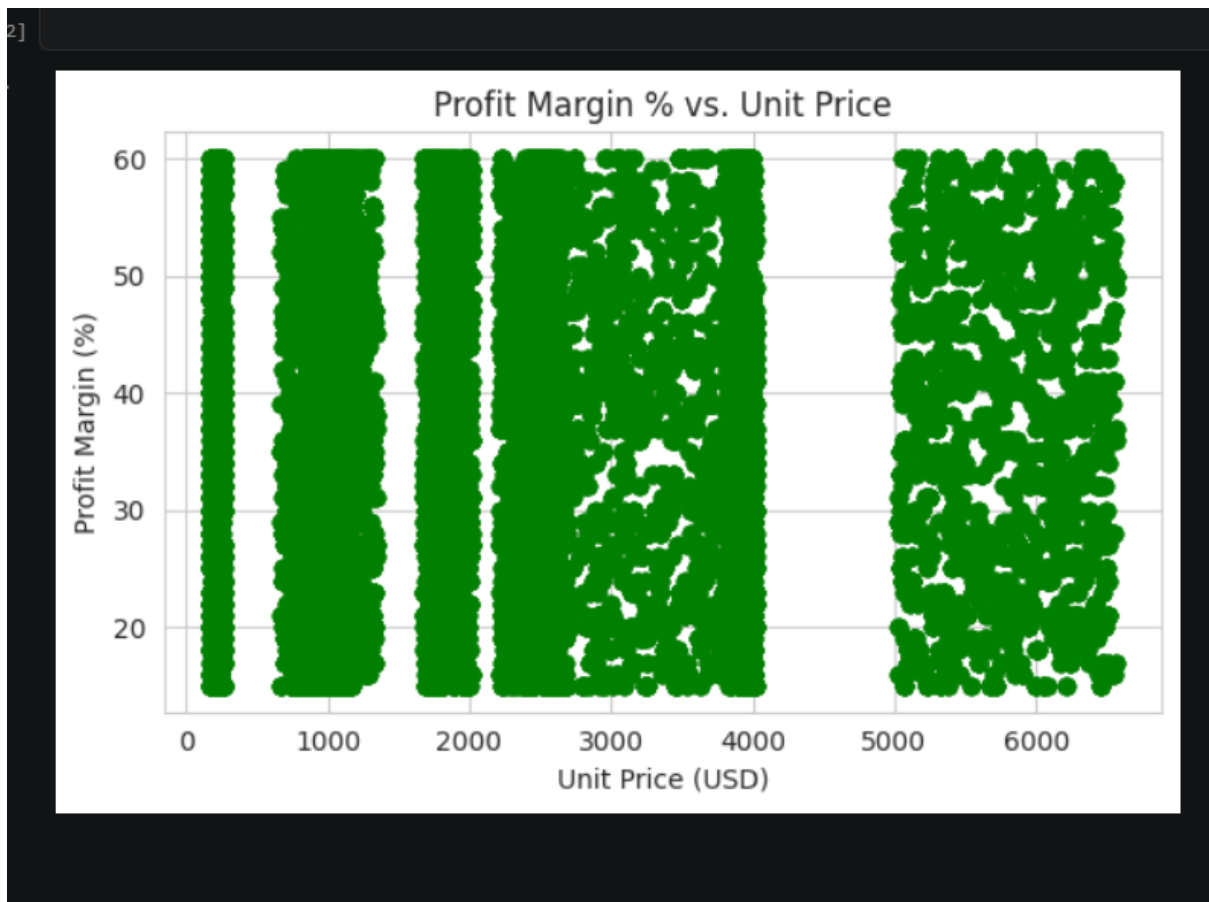


- **Top Products:** 26 & 25 dominate sales
- Mid-tier products show consistent performance
- Low-tier products need optimization

👉 Insight:

- Strong SKU concentration → risk + opportunity

📌 Profit vs Unit Price

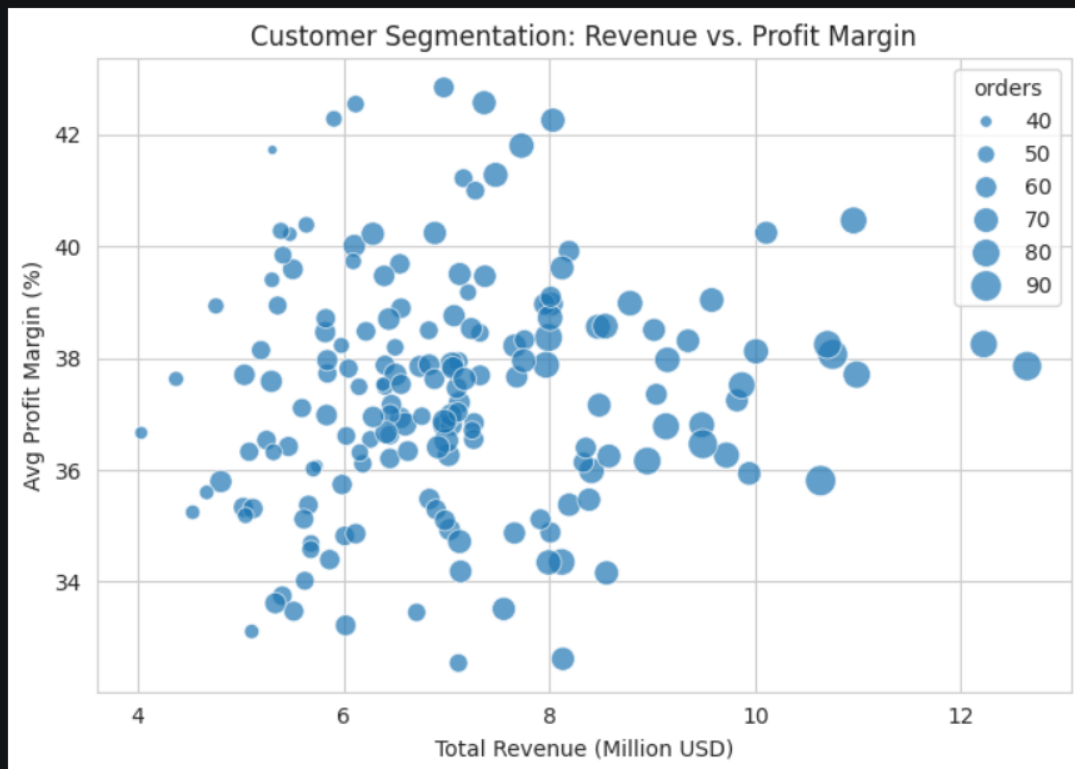


- Profit margin mostly between 30%–60%
- No strong relation with quantity

👉 Insight:

- Pricing (not volume) drives profitability

📌 Customer Segmentation

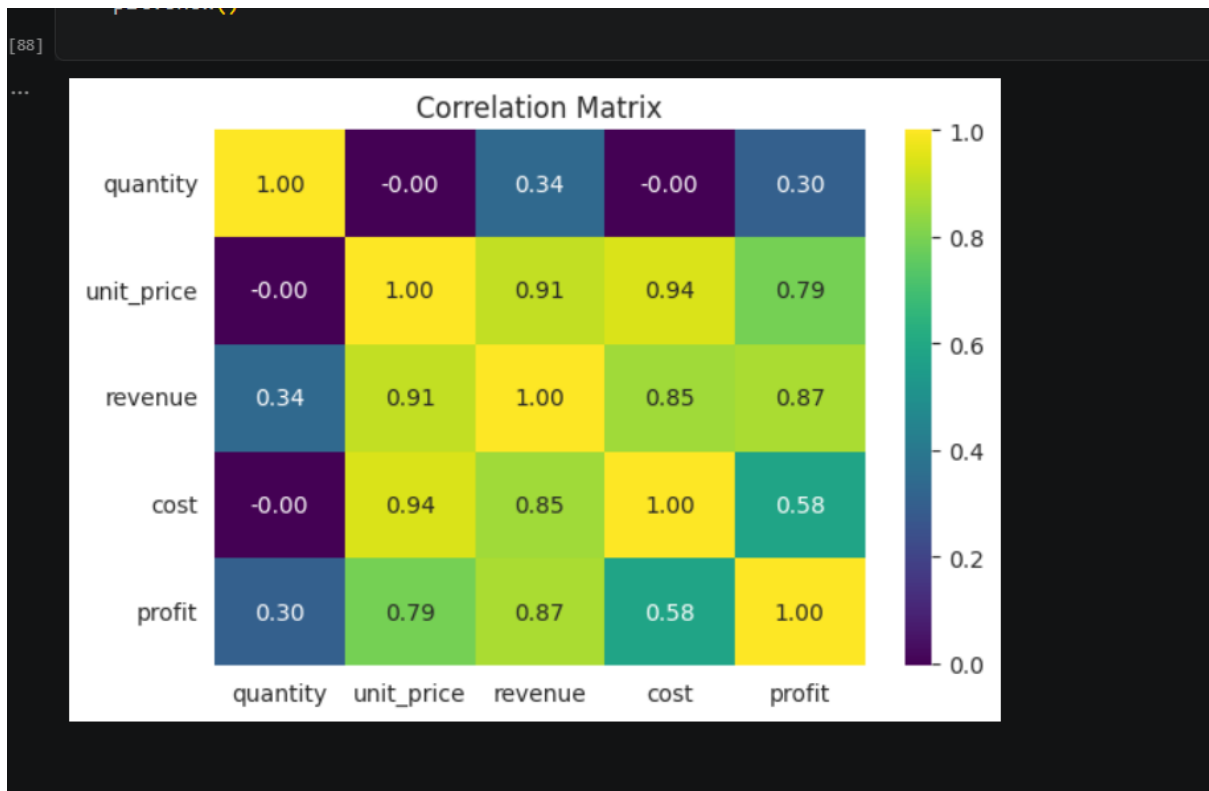


- High-revenue customers with lower margins → discounting issue
- Mid-revenue + high-margin customers → best targets

👉 Insight:

- Focus on high-margin segments for growth

📌 Correlation Analysis



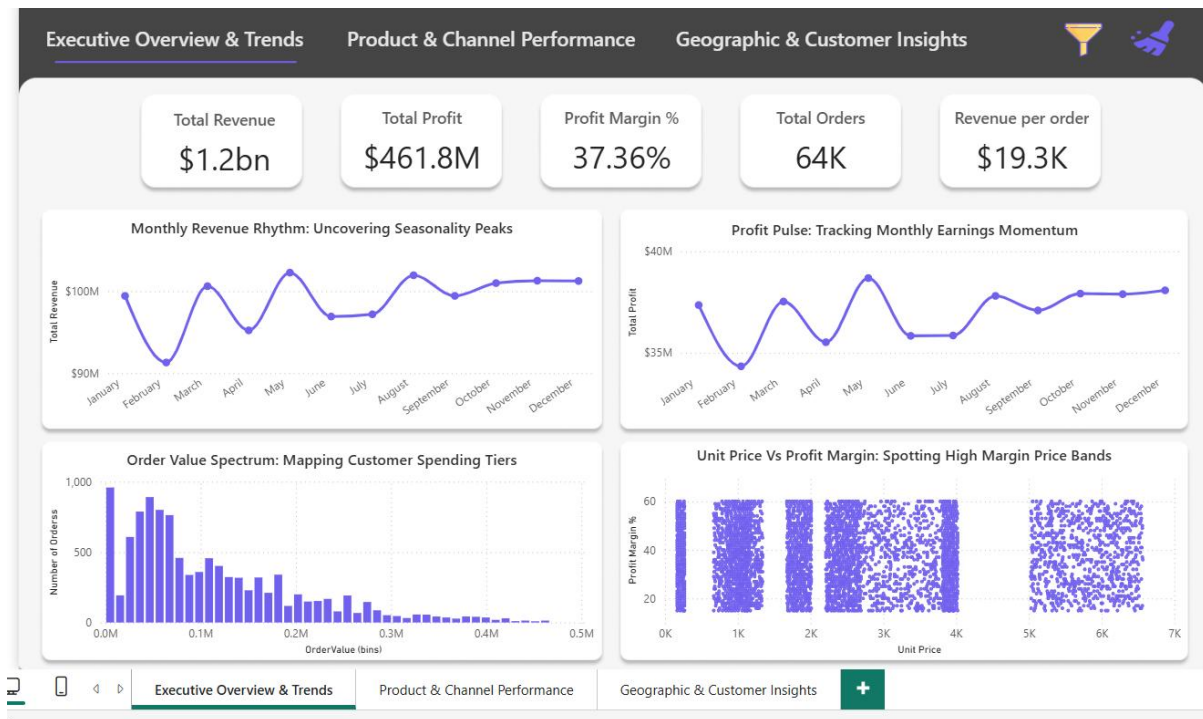
- **Unit Price strongly impacts:**
 - **Cost (0.94)**
 - **Revenue (0.91)**
 - **Profit (0.79)**
- **Quantity has weak impact**

👉 Insight:

- **Pricing strategy > Sales volume**

4. Dashboard Insights (Power BI)

Executive Overview

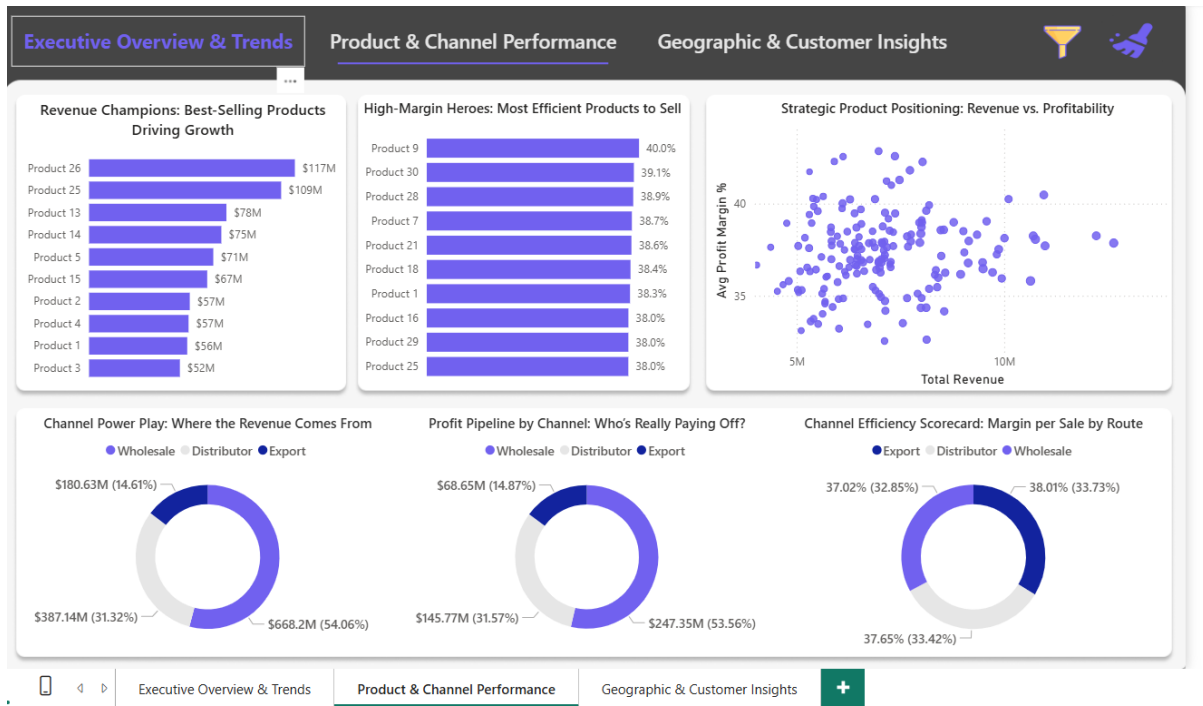


- Total Revenue: **\$1.2B**
- Profit: **\$461.8M**
- Profit Margin: **~37%**
- Orders: **64K**

👉 Seasonality:

- Peak: May–June
- Low: January

Product & Channel Performance

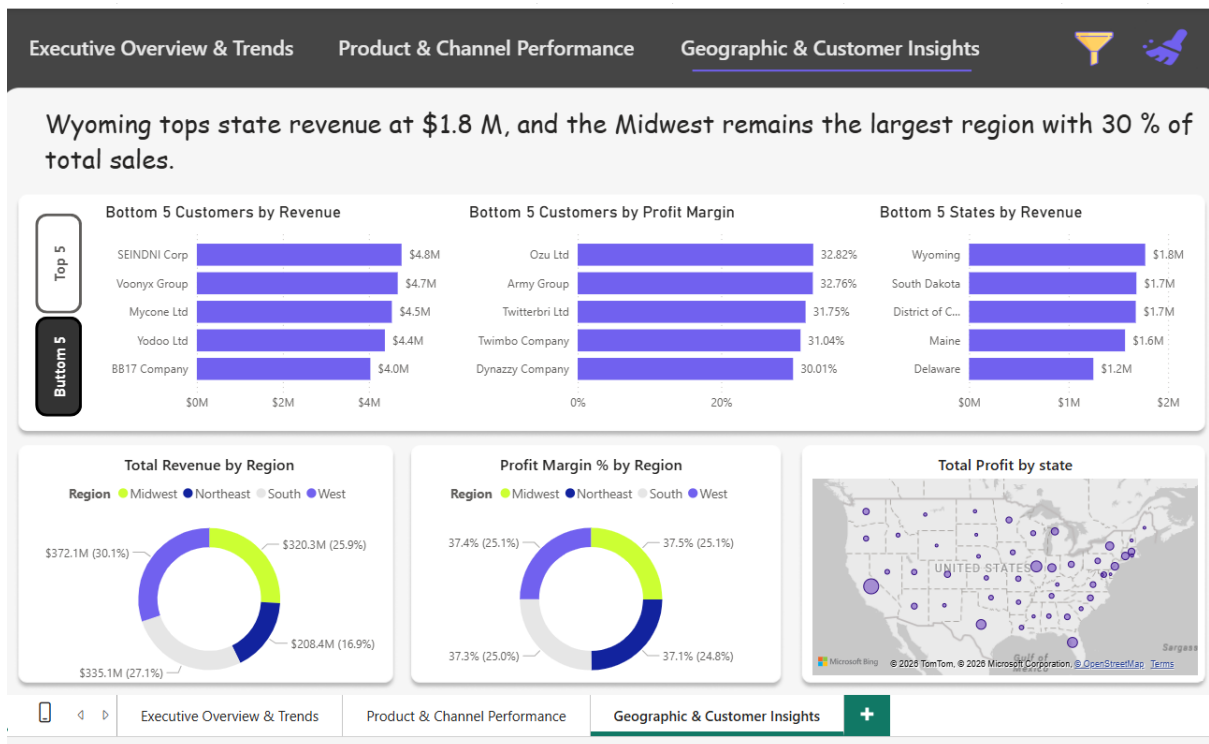


- Wholesale dominates (~54%)
- Export gives highest margin (~38%)
- Distributor moderate

👉 Insight:

- Volume vs Profit trade-off across channels

📌 Geographic Insights



- California leads in revenue
- West region strongest overall
- Northeast underperforming

👉 Insight:

- Regional imbalance → growth opportunity

🔥 5. Key Business Insights

- ✓ Strong **seasonality in sales**
- ✓ Revenue heavily depends on **few products & customers**
- ✓ **Pricing is the main driver** of profit
- ✓ Wholesale drives volume, Export drives profit
- ✓ California & West dominate sales
- ✓ Northeast has untapped potential

6. Recommendations

1. Seasonal Strategy

- Boost campaigns in **low months (Jan–Apr)**
 - Maximize peak months (May–June)
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2. Product Strategy

- Focus on **Top SKUs (26, 25)**
 - Improve or remove low-performing products
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3. Pricing Strategy

- Reduce unnecessary discounts
 - Optimize pricing for high-revenue customers
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4. Channel Strategy

- Expand **Export channel (high margin)**
 - Optimize Wholesale margins
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5. Regional Strategy

- Replicate California success
 - Invest in **Northeast & Midwest**
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6. Customer Strategy

- Target:
 - High-margin customers
 - Mid-tier high potential clients
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7. Conclusion

This project successfully:

- Delivered **end-to-end analytics (EDA → Dashboard)**
- Identified **key revenue & profit drivers**
- Enabled **data-driven decision making**

 Business impact:

- Better pricing decisions
- Improved product strategy
- Optimized regional expansion